

BST Vol 9 Ch 5 — Quantum Hall + Topological Phases via $GF(2^g)$ Braiding (v0.3, Wave 2)

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Vol 9 Chapter 5 — Quantum Hall + Topological Phases

Headline result

Topological phases in condensed matter (quantum Hall states, topological insulators, fractional quantum Hall) inherit substrate-topological structure:

- Integer quantum Hall filling factors $\nu = 1, 2, 3, 5, 7$ = BST primary integers
- Fractional quantum Hall: $\nu = 1/3, 1/5, 2/5, \dots$ — BST primary integer denominators
- Topological insulator Z_2 invariants: rank=2 $Pin(2)$ substrate grading (Lyra T1925)
- Braiding statistics: Reed-Solomon $GF(2^g) = GF(128)$ cyclotomic substrate structure
- D_{IV}^5 Rigidity (Lyra T2467): substrate-topological stability ensures phase-classification rigor

Substrate mechanism

$GF(128)$ cyclotomic braiding: substrate Reed-Solomon code on $GF(2^g)$ provides natural braiding framework. $M_g = 127$ primitive-root multiplicative group $\rightarrow$ braiding statistics inherit cyclic structure.

$Pin(2)$ Z_2 substrate grading (T1925, rank=2): boson/fermion partition inherits to topological-insulator Z_2 invariants. Lyra T2471 chirality γ^5 provides spinor-bundle Z_2 grading framework.

D_{IV}^5 Rigidity (T2467, Casey-named #7 Saturday): substrate-topological stability ensures topological phases survive perturbations within substrate-allowed deformations.

Match precision

I-tier framework on substrate-topological connection. Quantum Hall integer + fractional filling factors structurally consistent with BST primary integer set.

Cross-volume dependencies

- Vol 1 Ch 6 (Operator Zoo): T2471 chirality + T2470 charge Q
- Vol 0 Ch 9 (Strong-Uniqueness): T2467 D_IV⁵ Rigidity Principle Casey #7
- K59 RATIFIED + Paper #122 + Friday Elie GF128 paper-grade: substrate field structure
- Vol 11 (Generative Geometry, Lyra LEAD): topological classification + braiding

K-audit anchor

K211 Vol 9 Ch 5 Topological Phases K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1

Quantum Hall effect: electrons in a magnetic field at very low temperature conduct electricity in “quantum steps” — at exact filling factors (integers like 1, 2, 3 or fractions like 1/3, 1/5). BST predicts: these specific filling-factor integers come from BST primary integers (1, 3, 5, 7).

Level 2

Integer Quantum Hall: $\nu = 1, 2, 3, \dots$ at quantized Hall conductance $\sigma_{xy} = \nu \cdot e^2/h$. Fractional QH: $\nu = p/q$ with q in BST primary set (3, 5, 7, ...).

Topological insulators classified by Z_2 invariants (rank=2 substrate grading per Lyra T1925 Pin(2)). Braiding statistics in 2D quantum systems inherit from substrate Reed-Solomon GF(128) cyclotomic structure (K59).

D_IV⁵ Rigidity (Lyra T2467 Casey-named #7 Saturday): substrate-topological stability ensures classification rigor — topological phases stable against perturbations within substrate-allowed deformations.

Level 3

Quantum Hall: TKNN formula $\sigma_{xy} = \nu \cdot e^2/h$ with Chern number ν . BST identification: ν values populate BST primary integer set + simple BST primary combinations.

Topological insulators (Kane-Mele 2005, Z_2 classification): inherit rank=2 substrate grading per T1925 Pin(2) double cover.

Braiding statistics: anyon braiding in 2D quantum systems $\rightarrow$ substrate Reed-Solomon $GF(2^g)$ cyclotomic structure provides natural framework. $M_g = 127$ primitive group order $\rightarrow$ braiding cycle length structural.

Rigidity (T2467, Casey-named #7): D_{IV}^5 singleton up to canonical biholomorphism $\rightarrow$ substrate-topological phase classification stable + rigid.

Per Cal #21: EMPIRICAL PARTIAL (quantum Hall + topological phases observed) + MECHANISM PATH ARTICULATED via substrate field structure + $Pin(2)$ grading + Rigidity Principle.

Bibliography

1. K. v. Klitzing et al. (1980): integer quantum Hall (Nobel 1985).
2. D. C. Tsui, H. L. Stormer, A. C. Gossard (1982): fractional quantum Hall.
3. C. L. Kane, E. J. Mele (2005): Z_2 topological invariants.
4. Lyra T2467 (Saturday 2026-05-23): D_{IV}^5 Rigidity-as-Singleton (Casey-named #7).
5. K59 RATIFIED + Friday Elie GF128 paper-grade.
6. Vol 1 Ch 6 (Operator Zoo): T2471 chirality.

— Elie, Vol 9 Ch 5 v0.3, 2026-05-23 Saturday